Unique spanning-tree families bound the static response of every complete four-state Markov network

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Abstract

The steady-state sensitivity of a Markov network admits a geometric description in terms of families of directed spanning trees. Fancher and Horowitz conjectured that families uniquely determined by their combined directed-edge multiplicities suffice to bound every response. We prove this containment for the complete four-state network, for all twelve strictly positive finite transition rates and arbitrary signed logarithmic rate perturbations. The result is stronger than a separate convex decomposition for each perturbation: the entire matrix of rooted-tree edge marginals has one decomposition, with weights independent of the perturbation direction. A fixed rescaling by the stationary probabilities reduces arbitrary rates to balanced flows. After symmetric traffic is removed, every four-state residual flow belongs, up to relabeling, to one nonnegative tournament cone. Within this cone, 93 cubic coefficient matrices admit exact rational decompositions over the 346 unique families. We give the complete finite certificate and two separately implemented exact checkers. The construction makes the associated topological covariance response bounds unconditional for complete four-state networks, without detailed balance or a small-affinity assumption. It does not establish the conjecture for general graphs or the attainability of every point of the response hull.

1 Introduction

Transition rates determine how a stochastic kinetic network responds to a changing control parameter. Even in a small network, nonequilibrium circulation allows that response to depend on many rate ratios. Structural constraints can nevertheless restrict sensitivity without selecting a particular set of rates. The support bound of Owen and Horowitz [1], for example, limits sensitivity when a specified set of rates shares a multiplicative input. General static response formulas express changes in stationary probabilities and currents through reduced inverse matrices and projection operators [2]. A different exact structure is the affine dependence among steady currents when the controlled edges obey an appropriate connectivity condition [3].

Fancher and Horowitz [4] proposed a joint geometric description of stationary probability responses. A rooted spanning tree contributes the sum of the logarithmic rate perturbations along its edges. Choosing one tree for each root gives a tree family and an associated response vector. The matrix-tree theorem yields a convex representation using all families [their Eq. (5)]. They conjectured that one can retain only the *uniquely constructable* families: those determined uniquely by their full vector of directed-edge multiplicities [their Eq. (6)]. Their supplementary construction proves the triangle case, while their four-state tests provide finite numerical evidence. A proof for all positive four-state rates requires more than additional samples.

Here we establish that full four-state statement, together with a common decomposition of all rooted-tree edge marginals. The new step is a nonnegative coefficient certificate after reduction to a balanced-flow cone. The matrix-tree formula, vertex-dependent rescaling of outgoing rates,

and current conservation are established tools; the coefficient decomposition is the additional ingredient. Because the same weights work for every perturbation direction, the result controls the joint response rather than only a selected scalar sensitivity. It removes the conjectural premise of the covariance bounds in Ref. [4] for the complete four-state graph.

2 Model, tree families, and the result

Consider states $0, 1, 2, 3$ with one transition in each direction between every pair. With columns indexed by the source state, the master equation and its rates are

$$\dot{p} = W(\lambda)p, \qquad W_{ij}(\lambda) = k_{ij}e^{\lambda d_{ij}} \quad (i \neq j), \qquad W_{jj}(\lambda) = -\sum_{i\neq j} W_{ij}(\lambda). \tag{1}$$

All twelve k_{ij} are strictly positive and finite. We set $\sum_{i\neq j} k_{ij} = 1$ by choosing the time unit at $\lambda = 0$; no constraint on this sum is imposed as λ varies. The control parameter λ and its logarithmic rate slopes $d_{ij} \in [-1, 1]$ are dimensionless. Detailed balance is not assumed. Irreducibility gives a unique positive stationary vector $\pi(\lambda)$, normalized by $\mathbf{1}^T\pi(\lambda) = 1$. Write $\pi = \pi(0)$ and

$$r_j = \partial_\lambda \log \pi_j(\lambda)|_{\lambda=0}\,. \tag{2}$$

This is an exact first derivative of the stationary state at an arbitrary nonequilibrium base point. It is not a transient or finite-frequency response.

An inward tree rooted at j has one outgoing edge from each other state and a directed path from each state to j. Let $\mathcal{T}_j$ denote all such trees. For a directed edge $e = (i, j)$, the arrow is $j \to i$ and its rate is $k_e = k_{ij}$. Define

$$k^T = \prod_{e\in T} k_e, \qquad \tau_j(k) = \sum_{T\in\mathcal{T}_j} k^T, \qquad H_{je}(k) = \frac{\sum_{T\in\mathcal{T}_j : e\in T} k^T}{\tau_j(k)}. \tag{3}$$

Thus H is a 4×12 matrix of conditional edge-inclusion probabilities. The matrix-tree expression $\pi_j = \tau_j / \sum_\ell \tau_\ell$ gives

$$r = Hd - \mathbf{1}(\pi^T Hd). \tag{4}$$

A family $\alpha = (T_0^\alpha, T_1^\alpha, T_2^\alpha, T_3^\alpha)$ chooses one tree per root. Its incidence matrix and total directed-edge multiplicity are

$$I_{je}^\alpha = \mathbf{1}_{\{e\in T_j^\alpha\}}, \qquad m_e^\alpha = \sum_{j=0}^{3} I_{je}^\alpha. \tag{5}$$

Let $\mathcal{U}$ contain exactly those families for which no distinct family has the same twelve-component vector m^α. Opposite arrows are distinct coordinates. Each root has 16 trees, so there are $16^4 = 65\,536$ families. Full enumeration gives 5955 multiplicity classes, of which 346 are singletons. Appendix A fixes an explicit encoding and gives the complete witness used below.

Theorem 1 (Common marginal decomposition)**.** *For every strictly positive finite rate array k on the complete four-state graph, there are weights $\theta_\alpha(k) \geq 0$, $\alpha \in \mathcal{U}$, with $\sum_{\alpha\in\mathcal{U}} \theta_\alpha(k) = 1$ such that*

$$H(k) = \sum_{\alpha\in\mathcal{U}} \theta_\alpha(k) I^\alpha. \tag{6}$$

The weights depend only on the base-point rates. Consequently, for every $d \in [-1, 1]^{12}$, setting $G^\alpha = I^\alpha d$ gives

$$r = \sum_{\alpha\in\mathcal{U}} \theta_\alpha(k)\left[G^\alpha - \mathbf{1}(\pi^T G^\alpha)\right] \;\in\; \operatorname{conv}\{G^\alpha - \mathbf{1}(\pi^T G^\alpha) : \alpha \in \mathcal{U}\}. \tag{7}$$

The theorem concerns edge marginals, not a coupling of the complete probability laws on $\mathcal{T}_j$. It also makes no assertion that every point in the displayed hull is attained by finite positive rates. The proof below constructs one valid set of weights at each base point.

3 Reduction to one balanced-flow cone

3.1 A fixed stationary-flow rescaling

At the base point set

$$q_{ij} = k_{ij}\pi_j \quad (i \neq j). \tag{8}$$

Stationarity says that the incoming and outgoing q flows at every state are equal. Every tree rooted at j uses exactly one outgoing edge from each state $\ell \neq j$, so

$$q^T = \left(\prod_{\ell \neq j} \pi_\ell\right) k^T, \qquad H_{je}(q) = H_{je}(k). \tag{9}$$

The factor cancels from the conditional tree distribution. Such outgoing-rate rescalings are also discussed after Eq. (5) of Ref. [4]. The advantage of the particular choice (8) is that q, regarded as a rate array, has uniform stationary probabilities. Its four tree sums therefore agree: $\tau_j(q) = \tau(q)$.

Only the base-point array is made balanced. The rescaling in a perturbed family is $q_{ij}(\lambda) = \pi_j(0)k_{ij}e^{\lambda d_{ij}}$; the factor $\pi_j(0)$ is held fixed. The array $q(\lambda)$ need not remain balanced. Accordingly, all derivatives below are taken with respect to the twelve independent directed rates before imposing balanced-flow coordinates. The final centering in Eq. (7) uses the original π, not the uniform distribution of q.

3.2 Residual support and tournament classification

For every unordered pair put $s_{ij} = \min(q_{ij}, q_{ji}) > 0$ and subtract that amount from both directions. The residual is a nonnegative balanced flow with at most one positive direction on each pair. It has no positive edge between distinct strongly connected components. Indeed, a source component in the acyclic condensation graph has zero incoming flow, so conservation of its total flow forces zero outgoing flow. Removing that component and repeating proves the claim.

A nontrivial residual component contains at least three vertices, since an opposite pair cannot remain. Thus the nonzero residual support on four vertices is either strongly connected on all four vertices or a directed triangle plus an isolated vertex. If it is strongly connected, orient missing pairs to complete a strongly connected tournament. For a triangle, give the fourth vertex at least one incoming and one outgoing arrow. The completed tournament is again strongly connected. If the residual is zero, choose any strongly connected tournament. Added arrows carry zero residual flow and do not remove any original positive rate.

There is only one isomorphism class of strongly connected four-vertex tournaments. Every vertex has outdegree one or two, and the sum of the outdegrees is six. Let the two outdegree-two vertices be a, b, with $a \to b$. Vertex a beats one low-degree vertex c and loses to the other, d. Vertex b must beat both c and d, and c must beat d. These choices determine all arrows. We use the representative

$$0 \to 1, \quad 1 \to 2, \quad 2 \to 0, \quad 0 \to 3, \quad 3 \to 1, \quad 2 \to 3. \tag{10}$$

Let $A, B, C \geq 0$ be the residual flows on $0 \to 1$, $2 \to 3$, and $0 \to 3$, respectively. Conservation then fixes the residual flows on $2 \to 0$, $3 \to 1$, and $1 \to 2$ to $A + C$, $B + C$, and $A + B + C$. They

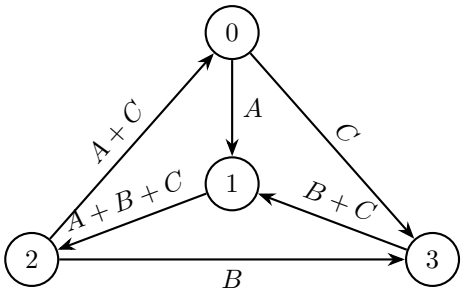


Figure 1: Canonical residual flow after removal of symmetric traffic. Arrow labels are flow magnitudes, with $A, B, C \geq 0$. To recover the strictly positive rate array q, add $s_{ij} > 0$ in both directions on each of the six pairs. Every balanced positive four-state array has this form after a state permutation; an arrow may have zero residual flow while both original rates remain positive.

are equivalently the coefficients of the three directed cycles $0 \to 1 \to 2 \to 0$, $1 \to 2 \to 3 \to 1$, and $0 \to 3 \to 1 \to 2 \to 0$ (Fig. 1).

Thus every positive balanced array belongs, after relabeling, to the following cone:

$\{i,j\}$	q_{ij} $(i<j)$	q_{ji} $(i<j)$
01	s_{01}	$s_{01} + A$
02	$s_{02} + A + C$	s_{02}
03	s_{03}	$s_{03} + C$
12	s_{12}	$s_{12} + A + B + C$
13	$s_{13} + B + C$	s_{13}
23	s_{23}	$s_{23} + B$

$$s_{ij} > 0, \quad A, B, C \geq 0. \tag{11}$$

This is a covering construction for arbitrary rates, including tied opposite flows; it imposes no symmetry of the six s_{ij}. State permutations act bijectively on families and on full multiplicity classes, preserving membership in $\mathcal{U}$.

4 The finite coefficient certificate

Let $x = (s_{01}, s_{02}, s_{03}, s_{12}, s_{13}, s_{23}, A, B, C)$. After substituting Eq. (11), write

$$\tau_j(q(x)) = \tau(x) = \sum_{m \in \mathcal{M}} c_m x^m, \tag{12}$$

$$N_{je}(x) := \left[q_e \frac{\partial \tau_j(q)}{\partial q_e} \right]_{q=q(x)} = \sum_{m \in \mathcal{M}} N_{m,je} x^m. \tag{13}$$

The derivative in (13) marks an original directed edge, not a cone variable. Every tree has three edges, so these polynomials have degree exactly three. Direct expansion gives 93 nonzero monomials: 82 have $c_m = 1$, ten have $c_m = 2$, and ABC has $c_m = 3$. All coefficients are nonnegative integers. The support of each numerator is contained in that of τ because it sums a subset of the rooted tree terms.

Lemma 1 (Coefficient decomposition)**.** *For every $m \in \mathcal{M}$, there are nonnegative rational numbers $\beta_{m,\alpha}$ supported on unique families such that*

$$\sum_{\alpha \in \mathcal{U}} \beta_{m,\alpha} = 1, \qquad N_{m,je} = c_m \sum_{\alpha \in \mathcal{U}} \beta_{m,\alpha} I^{\alpha}_{je} \quad \text{for all } j, e. \tag{14}$$

The complete witness in Table 2 uses c_m families for each monomial, each with weight $1/c_m$.

Finite exact verification. The edge order and tree table in Appendix A decode each listed family into its four trees. Enumerating all choices of one tree per root and grouping by the full twelve-dimensional multiplicity vector establishes the singleton status of every witness family. Expanding each tree product with Eq. (11) gives both c_m and the marked-edge coefficient matrix N_m. For each row of Table 2, adding the listed family incidence matrices gives exactly N_m, since the factor c_m cancels the weight $1/c_m$. This verifies all $93 \times 4 \times 12 = 4464$ component identities in (14), together with positivity and normalization.

The accompanying programs perform this full finite calculation in integer and rational arithmetic. One reconstructs trees by parent maps and the polynomials from the 3×3 principal cofactors of the column Laplacian $-W$. The other generates trees from Prüfer sequences and expands products by multisets of symmetric-flow and cycle labels. Both regenerate all families and all nonzero monomials; neither assumes the stored numerator matrices or singleton classifications are correct. Only the proposed supports and rational weights are treated as witnesses to be checked. Appendix B states the verification procedure and package interface. There is no floating tolerance or truncation in these identities. □

4.1 The coefficient of ABC

The only coefficient equal to three illustrates both the certificate and its scope. In the directed-edge order

$$E = (01, 02, 03, 10, 12, 13, 20, 21, 23, 30, 31, 32), \qquad ij \equiv j \to i, \tag{15}$$

the marked-edge coefficient is

$$N_{ABC} = \begin{pmatrix} 0&3&0&0&0&3&0&3&0&0&0&0 \\ 0&2&0&2&0&3&0&0&0&1&0&1 \\ 0&0&0&2&0&3&0&3&0&1&0&0 \\ 0&1&0&1&0&0&0&3&0&2&0&2 \end{pmatrix} = I^{41333} + I^{41336} + I^{44990}. \tag{16}$$

For root 0, the residual support has the single tree $\{2 \to 0, 3 \to 1, 1 \to 2\}$, whose product $(A+C)(B+C)(A+B+C)$ has ABC coefficient three. For root 1, the four possible tree products are

$$A(A+C)(B+C), \quad AB(B+C), \quad C(A+C)(B+C), \quad CB(B+C). \tag{17}$$

Their ABC coefficients are $1, 1, 1, 0$. The certificate uses twice the tree associated with the first product and once the tree associated with the fourth. These choices preserve every edge marginal, as (16) shows, but change the full tree distribution of this coefficient. The construction therefore must not be interpreted as reproducing the complete rooted-tree laws. That stronger interpretation is unnecessary for G_j^α, which is linear in the edge indicators. This example does not exclude a different full-law construction.

5 Constructing the weights and bounding response

For any cone coordinates representing a positive balanced flow, define

$$\rho_m(x) = \frac{c_m x^m}{\tau(x)}, \qquad \theta_\alpha(x) = \sum_{m \in \mathcal{M}} \rho_m(x) \beta_{m,\alpha}. \tag{18}$$

All ρ_m are nonnegative and their sum is one. The denominator is strictly positive, since all original q_{ij} are positive; equivalently, the positive symmetric traffic already supplies positive tree products. Hence all θ_α are nonnegative and normalized, even when one or more of A, B, C vanish. Dividing (14) by $\tau(x)$ and summing with x^m gives

$$H_{je}(q) = \frac{N_{je}(x)}{\tau(x)} = \sum_{\alpha \in \mathcal{U}} \theta_\alpha(x) I_{je}^\alpha. \tag{19}$$

Transport these family labels back through the state permutation used in Sec. 3. Equation (9) proves (6) for the original k. Multiplication by d and subtraction of the original π mean in (4) prove (7). This completes the proof of Theorem 1.

No exchange of a singular rate limit is involved. A choice of tournament can change at a zero residual-current boundary, but the theorem requires existence of weights at each base point, not a globally smooth selection. In a chosen canonical cone the displayed certificate uses 78 distinct unique families; the full set of 346 is retained in the invariant statement. Under a common rate rescaling, all cubic monomials and τ acquire the same factor, leaving the weights unchanged.

For a fixed state observable $Q = (Q_0, Q_1, Q_2, Q_3)$, let $\langle Q\rangle_\pi = \sum_j \pi_j Q_j$ and $\mathrm{Cov}_\pi(Q, G) = \langle QG\rangle_\pi - \langle Q\rangle_\pi \langle G\rangle_\pi$. The observable itself is assumed independent of λ. Then

$$\partial_\lambda \langle Q\rangle_{\pi(\lambda)}\Big|_0 = \sum_{\alpha\in\mathcal{U}} \theta_\alpha(k)\, \mathrm{Cov}_\pi(Q, G^\alpha), \tag{20}$$

$$\min_{\alpha\in\mathcal{U}} \mathrm{Cov}_\pi(Q, G^\alpha) \le \partial_\lambda \langle Q\rangle_{\pi(\lambda)}\Big|_0 \le \max_{\alpha\in\mathcal{U}} \mathrm{Cov}_\pi(Q, G^\alpha). \tag{21}$$

The bound formula is the specialization of Eq. (8) of Ref. [4]; the contribution here is a proof of its underlying unique-family containment for all positive complete four-state networks. At a specified base point the extrema require a finite comparison over the 346 families. Their values still depend on π, Q, and d; the theorem does not eliminate that physical dependence or assert finite-rate saturation of an extremum.

The dimensions are consistent throughout: k, q, s, and the cycle flows have units of inverse time; τ and N have units of inverse time cubed. The quantities H, ρ, θ, r, d, and λ are dimensionless. Equation (21) consequently has the units of Q. This finite-state model has no spatial boundary or infinite-dimensional tail to truncate.

6 Verification, comparison, and limitations

The universal rate quantifier is carried by the analytic reduction and the complete finite polynomial identities. Additional exact rate tests check the convention map and the implementation. The Prüfer checker reconstructs 210 positive-rate cases, including all 24 state permutations for each zero/nonzero pattern of A, B, C, nonuniform stationary probabilities, and a finite rate ratio 10^{36}. For every case it compares the complete marginal matrix and the twelve independent response directions against exact row-generator linear solves. It also enumerates all 64 tournaments and all $3^6 = 729$ residual directional supports; all 75 possible positive supports of balanced flows are contained in the 24 relabelings of (10). These finite checks supplement the analytic coverage argument; no sample count is used as evidence for an untested continuum of rates.

The published two-state single-binding specialization of Ref. [1], $W_{10} = x$ and $W_{01} = 1$, provides a simple normalization check. Direct solution gives $\pi_1 = x/(1+x)$, $\partial_{\log x}\pi_1|_{x=1} = 1/4$, and effective Hill coefficient one. The triangle family counts are also recovered: 27 families and 11 singletons, as in Ref. [4]. These are reproduced baselines, not additional results of this paper.

The general inverse/projection formulas of Ref. [2], Eqs. (6)–(8) of its cited preprint version, compute static response but do not supply the nonnegative unique-family coefficients used here. The affine current relation of Ref. [3], Eq. (7) of its cited preprint version, assumes that deleting the controlled input edges leaves a connected graph; its coefficients are not specified as nonnegative normalized weights. It is a different statement from the simultaneous directed-rate probability-response decomposition (6). These comparisons identify the concrete additional content relative to the displayed prior formulas, without claiming an exhaustive exclusion of every possible indirect derivation.

The proof is limited to the complete four-state graph with all twelve rates strictly positive and finite. It does not resolve the general-graph conjecture, establish full-hull attainability, or

impose a complete tree-law coupling. It concerns first-order static response at an arbitrary base point, with no claim about higher derivatives or finite perturbation errors. The result describes a stochastic model rather than an experimentally identified biochemical mechanism. Within this scope, the common convex weights give a constructive bridge from continuous rate parameters to the finite unique-family response bounds.

Data and code availability

The full rational certificate is included as `supplement/certificate.json`; its complete mathematical content is tabulated in Appendix A. The accompanying `check_cofactors.py` and `check_prufer.py` programs reconstruct and verify the certificate using only the Python standard library. The supplement provides execution instructions, conventions, environment information, and table-regeneration code. No external dataset, proprietary solver, or downloaded third-party code is needed to check the finite identities.

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A Complete certificate and decoding conventions

Number the twelve directed edges from 0 to 11 in the order (15). At each root, sort the three edge indices within each inward tree, then sort the resulting triples lexicographically. The tree index t runs from 0 to 15. Table 1 lists all these triples. A family's integer ID is

$$f = 16^3 t_0 + 16^2 t_1 + 16 t_2 + t_3. \tag{22}$$

The root labels are part of this definition: exchanging two trees between roots is not an allowed identification. Uniqueness is checked against all 65 536 IDs using their full directed-edge multiplicities.

Table 1: Complete rooted-tree encoding. Each entry is a triple of zero-based directed-edge indices, not a triple of vertex labels.

t	Root 0	Root 1	Root 2	Root 3
0	0, 1, 2	1, 2, 3	0, 2, 6	0, 1, 9
1	0, 1, 5	1, 3, 5	0, 5, 6	0, 4, 9
2	0, 1, 8	1, 3, 8	0, 6, 8	0, 6, 11
3	0, 2, 4	1, 5, 9	0, 8, 9	0, 9, 11
4	0, 2, 11	2, 3, 4	2, 3, 7	1, 3, 10
5	0, 4, 5	2, 3, 11	2, 6, 7	1, 7, 9
6	0, 4, 8	2, 4, 6	2, 6, 10	1, 9, 10
7	0, 5, 11	3, 4, 5	3, 5, 7	3, 4, 10
8	1, 2, 7	3, 4, 8	3, 7, 8	3, 7, 11
9	1, 2, 10	3, 5, 11	3, 8, 10	3, 10, 11
10	1, 5, 7	4, 5, 6	5, 6, 7	4, 6, 10
11	1, 7, 8	4, 5, 9	5, 7, 9	4, 9, 10
12	1, 8, 10	4, 6, 8	6, 7, 8	6, 7, 11
13	2, 4, 10	4, 8, 9	6, 8, 10	6, 10, 11
14	2, 7, 11	5, 6, 11	7, 8, 9	7, 9, 11
15	2, 10, 11	5, 9, 11	8, 9, 10	9, 10, 11

Table 2 specifies every nonzero monomial in (12), its scalar coefficient, and the unique-family support of its convex coefficient decomposition. All unlisted cubic monomials have zero coefficient. Each listed family has weight $1/c_m$, and all other $\beta_{m,\alpha}$ are zero. Thus the table and the tree decoder specify the full finite witness without requiring a particular numerical solver or an omitted coefficient file.

Table 2: Complete coefficient witness. Every listed family receives weight $1/c_m$; unlisted families receive zero. IDs use Eq. (22).

x^m	c_m	Family IDs	Each weight
C^3	1	41909	1
$B\,C^2$	2	41909, 44990	1/2
$B^2\,C$	1	44990	1
$A\,C^2$	2	41333, 41909	1/2
$A\,B\,C$	3	41333, 41336, 44990	1/3
$A\,B^2$	1	43384	1
$A^2\,C$	1	41333	1
$A^2\,B$	1	41336	1

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Table 2 continued

x^m	c_m	Family IDs	Each weight
$s_{23}\,C^2$	1	49134	1
$s_{23}\,B\,C$	1	49134	1
$s_{23}\,A\,C$	2	45806, 47496	1/2
$s_{23}\,A\,B$	1	47496	1
$s_{23}\,A^2$	1	45704	1
$s_{13}\,C^2$	1	41910	1
$s_{13}\,B\,C$	1	44991	1
$s_{13}\,A\,C$	2	41332, 41910	1/2
$s_{13}\,A\,B$	1	43385	1
$s_{13}\,A^2$	1	41332	1
$s_{13}\,s_{23}\,C$	1	53247	1
$s_{13}\,s_{23}\,A$	1	51609	1
$s_{12}\,C^2$	1	43957	1
$s_{12}\,B\,C$	1	43966	1
$s_{12}\,A\,C$	1	42869	1
$s_{12}\,A\,B$	1	42872	1
$s_{12}\,s_{23}\,C$	1	48622	1
$s_{12}\,s_{23}\,A$	1	47240	1
$s_{12}\,s_{13}\,C$	1	43963	1
$s_{12}\,s_{13}\,A$	1	42871	1
$s_{03}\,C^2$	1	33717	1
$s_{03}\,B\,C$	2	33717, 61374	1/2
$s_{03}\,B^2$	1	61374	1
$s_{03}\,A\,C$	2	32837, 33717	1/2
$s_{03}\,A\,B$	2	33717, 58702	1/2
$s_{03}\,A^2$	1	32837	1
$s_{03}\,s_{23}\,C$	1	61422	1
$s_{03}\,s_{23}\,B$	1	61422	1
$s_{03}\,s_{23}\,A$	1	58862	1
$s_{03}\,s_{13}\,C$	1	37814	1
$s_{03}\,s_{13}\,B$	1	65471	1
$s_{03}\,s_{13}\,A$	1	37814	1
$s_{03}\,s_{13}\,s_{23}$	1	65535	1
$s_{03}\,s_{12}\,C$	1	35765	1
$s_{03}\,s_{12}\,B$	1	60350	1
$s_{03}\,s_{12}\,A$	1	33861	1
$s_{03}\,s_{12}\,s_{23}$	1	60910	1
$s_{03}\,s_{12}\,s_{13}$	1	56251	1
$s_{02}\,C^2$	1	41893	1
$s_{02}\,B\,C$	2	41893, 44716	1/2
$s_{02}\,B^2$	1	44716	1
$s_{02}\,A\,C$	1	41381	1
$s_{02}\,A\,B$	1	41388	1
$s_{02}\,s_{23}\,C$	1	48844	1
$s_{02}\,s_{23}\,B$	1	48844	1
$s_{02}\,s_{23}\,A$	1	45772	1
$s_{02}\,s_{13}\,C$	1	41894	1
$s_{02}\,s_{13}\,B$	1	44717	1
$s_{02}\,s_{13}\,A$	1	41380	1
$s_{02}\,s_{13}\,s_{23}$	1	52957	1
$s_{02}\,s_{12}\,C$	1	43685	1
$s_{02}\,s_{12}\,B$	1	43692	1
$s_{02}\,s_{12}\,s_{23}$	1	48332	1
$s_{02}\,s_{12}\,s_{13}$	1	43690	1

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Table 2 continued

x^m	c_m	Family IDs	Each weight
$s_{02}\, s_{03}\, C$	1	33621	1
$s_{02}\, s_{03}\, B$	1	33621	1
$s_{02}\, s_{03}\, A$	1	32853	1
$s_{02}\, s_{03}\, s_{13}$	1	37734	1
$s_{02}\, s_{03}\, s_{12}$	1	34389	1
$s_{01}\, C^2$	1	4464	1
$s_{01}\, B\, C$	2	4472, 31091	1/2
$s_{01}\, B^2$	1	31096	1
$s_{01}\, A\, C$	1	4464	1
$s_{01}\, A\, B$	1	4472	1
$s_{01}\, s_{23}\, C$	2	8840, 31027	1/2
$s_{01}\, s_{23}\, B$	1	31112	1
$s_{01}\, s_{23}\, A$	1	8840	1
$s_{01}\, s_{13}\, C$	1	4468	1
$s_{01}\, s_{13}\, B$	1	31097	1
$s_{01}\, s_{13}\, A$	1	4468	1
$s_{01}\, s_{13}\, s_{23}$	1	31129	1
$s_{01}\, s_{12}\, C$	1	22385	1
$s_{01}\, s_{12}\, B$	1	22392	1
$s_{01}\, s_{12}\, s_{23}$	1	26760	1
$s_{01}\, s_{12}\, s_{13}$	1	22391	1
$s_{01}\, s_{03}\, C$	1	64	1
$s_{01}\, s_{03}\, B$	1	17731	1
$s_{01}\, s_{03}\, A$	1	64	1
$s_{01}\, s_{03}\, s_{23}$	1	17715	1
$s_{01}\, s_{03}\, s_{12}$	1	13377	1
$s_{01}\, s_{02}\, C$	1	4368	1
$s_{01}\, s_{02}\, B$	1	4370	1
$s_{01}\, s_{02}\, s_{23}$	1	8738	1
$s_{01}\, s_{02}\, s_{13}$	1	4372	1
$s_{01}\, s_{02}\, s_{03}$	1	0	1

B Exact checking and construction procedure

The following finite procedure verifies the coefficient lemma.

1. Generate every inward tree at each root and order them as in Table 1. Enumerate all four-tuples, compute their twelve directed-edge multiplicities, and retain the singleton classes.

2. Construct each τ_j either as the sum of its tree products or as a principal cofactor of $-W$. Form $q_e \partial_{q_e} \tau_j$ in twelve independent variables before applying (11).

3. Expand all polynomials exactly in the nine cone variables. Check equality of the four denominators, their complete monomial support, and all scalar coefficients in Table 2.

4. Decode every witness ID, check singleton membership and rational weight normalization, and compare all 48 entries of (14) for each of the 93 monomials.

The two packaged checkers use different tree and polynomial representations for these steps. Both use integer polynomial coefficients and `fractions.Fraction`; floating-point linear programming was used only to discover candidate supports, which were then rationally verified. The exact degree-three bound follows from three-edge trees, not a fitted degree estimate.

To construct the response weights for a specified positive array, solve $W\pi = 0$ with $\mathbf{1}^T\pi = 1$, set $q_{ij} = k_{ij}\pi_j$, and find a state permutation whose canonical tournament contains every positive residual arrow. At most 24 permutations need to be considered. Read s_{ij} and A, B, C from (11), evaluate (18), and map each family back to the original state labels. This construction uses no optimization over d.

The supplied checkers accept no numerical tolerance and terminate with an error if a checked identity fails. They read the certificate relative to their own location and write their result files in the current working directory. Python 3.9.6 was used for the recorded checks; only standard-library modules are required. The supplementary README gives commands that can be run from an empty output directory. The analytic rescaling and cone-coverage arguments remain part of the mathematical proof and must be read together with the finite verification.

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